

EcoTrack

Satellite-based estimation of urban fossil-fuel CO₂ emissions
Shivajinagar → Talegaon Dabhade corridor, Pune

Progress report: what has been done, what was found, and what comes next

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Headline so far: Pune + Pimpri-Chinchwad emit about **2.79 million tonnes of fossil CO₂ per year** (annual mean; 95% range 1.79–4.36), estimated from satellite NO₂, CO and wind data. One of the two official inventories (EDGAR, 3.6) agrees; the other (ODIAC, 11.8) is ruled out.

1. The problem and the idea

Cities produce most fossil CO₂, but the official numbers (emission *inventories*) are slow, coarse and often disagree. For Pune, two respected inventories differ by a factor of 3.2. Measuring CO₂ directly from space is hard: Pune adds only ~0.1 ppm to a ~420 ppm background, far below the noise of today's CO₂ satellites.

The idea (from the base paper, Xie et al. 2026): burning fuel releases CO₂ *and* NO₂. NO₂ is easy to see from space (low background, lives only ~1–2 hours, measured daily by TROPOMI). So measure the city's NO₂ emission, then convert it to CO₂ with a CO₂:NOx ratio. That ratio is the weak link, and the base paper's largest error term.

Research question: can combining several satellites (NO₂ + CO + CO₂), weather and human-activity data estimate urban CO₂ better than relying on NO₂ alone?

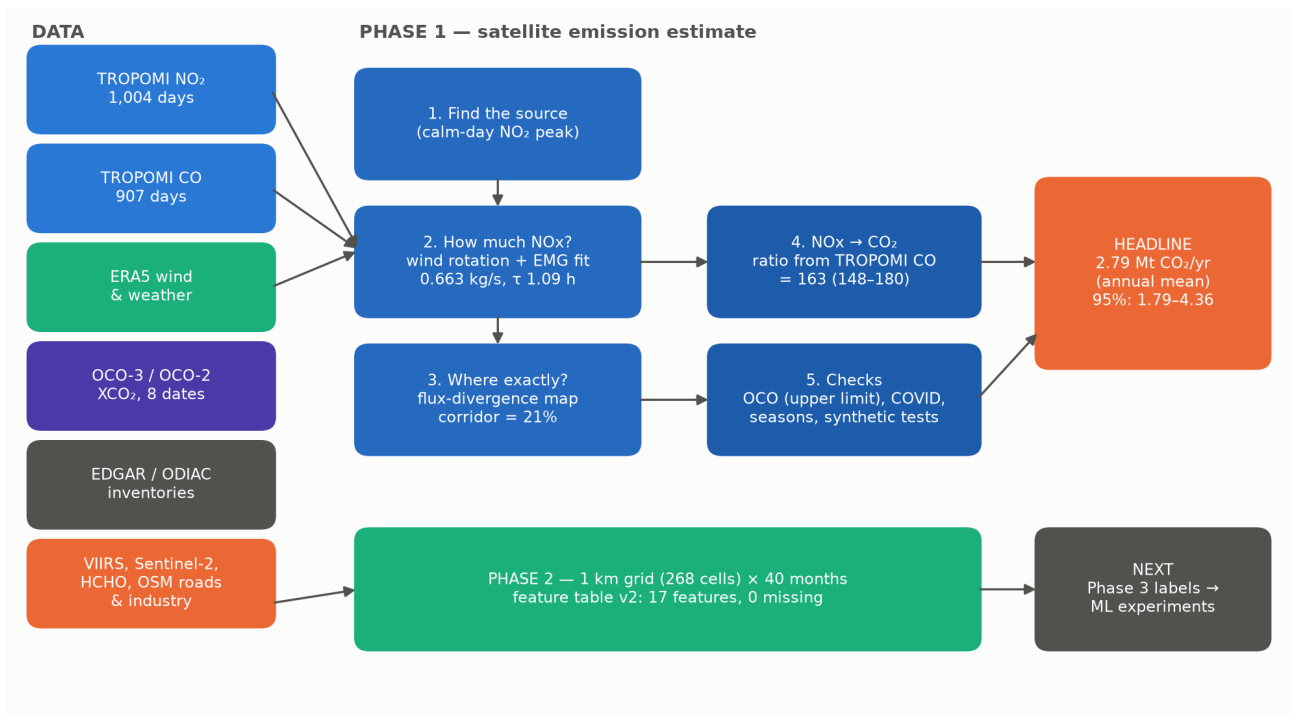


Figure 1. The whole workflow so far: data → Phase 1 emission estimate → checks → headline; Phase 2 builds the 1 km feature table for the machine-learning phases.

2. Study area and data

A 33 km strip along the Old Mumbai–Pune Highway where factories (MIDC Bhosari, Chinchwad, Talegaon) and heavy traffic sit side by side. The corridor is the highway centreline ± 3.5 km plus Talegaon MIDC (269 km²), divided into 268 grid cells of 1 km and three clusters.

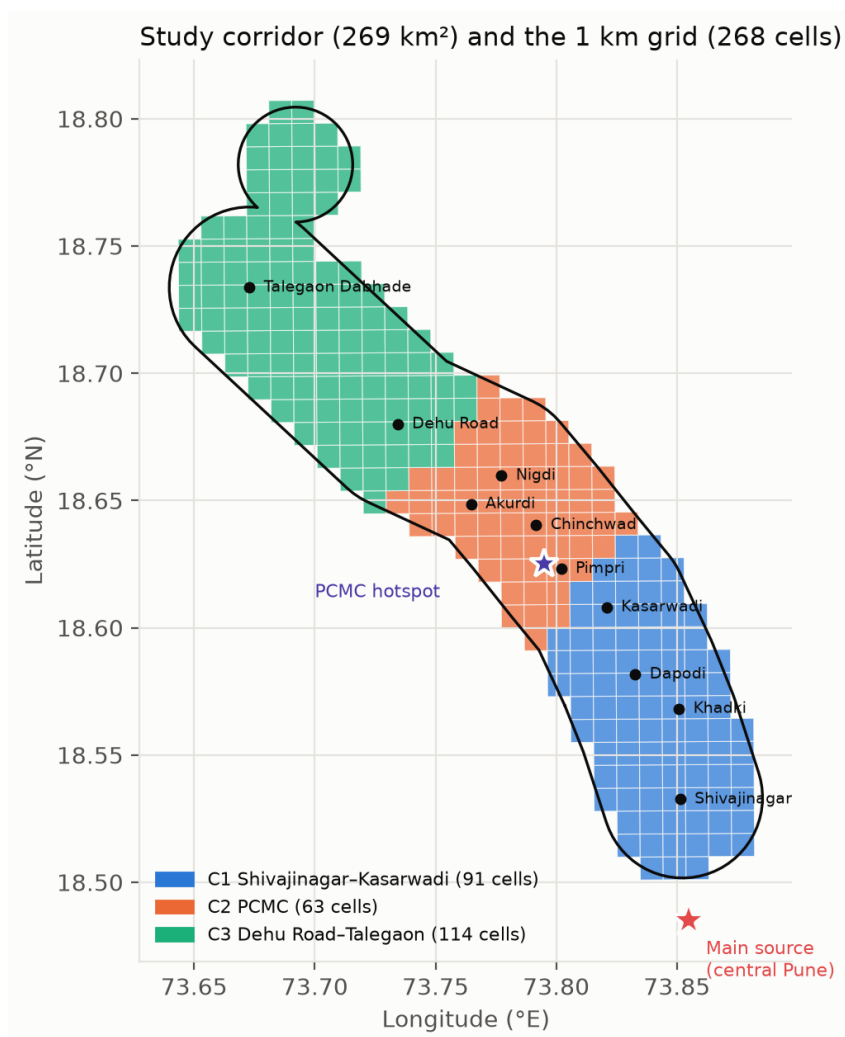


Figure 2. The corridor, its 1 km grid coloured by cluster, the ten waypoints, the main emission source found in Phase 1 (central Pune, just south of the corridor) and the second hotspot (Pimpri–Chinchwad).

Dataset	What it measures	Used for
TROPOMI (Sentinel-5P)	NO ₂ , CO, HCHO columns; daily, $\sim 3.5 \times 5.5$ km	Emissions, sector mix, chemistry proxy
OCO-3 / OCO-2 (NASA)	Column CO ₂ (XCO ₂); 8 usable dates over Pune	Direct CO ₂ check
ERA5 (ECMWF)	Hourly wind, boundary layer, temperature, sunlight	Moving the plumes; features
EDGAR, ODIAC	Bottom-up emission inventories	CO ₂ :NO _x prior, comparison (never truth)
VIIRS, Sentinel-2	Night lights; NDVI/NDBI	Human-activity features
OpenStreetMap / GRIP4	Roads, places, industrial land	Corridor geometry, road density
SRTM	Ground height	Terrain correction for CO

3. First: checking the data exist (feasibility, G1–G4)

Before building anything, four go/no-go checks with pass rules set in advance:

Check	Result	Verdict
G2 TROPOMI coverage	Median 26 usable days per month; worst month 13	PASS
G3 NO ₂ signal	All clusters 1.7–2.4× above clean background, every season	PASS
G1 OCO-3 / OCO-2	Pune is an OCO-3 'snapshot' target; 8 strong dates	Case study only
G4 Waypoints	6 of 10 were > 500 m off (Dehu Road 4.4 km); corridor rebuilt on the real highway	FIXED

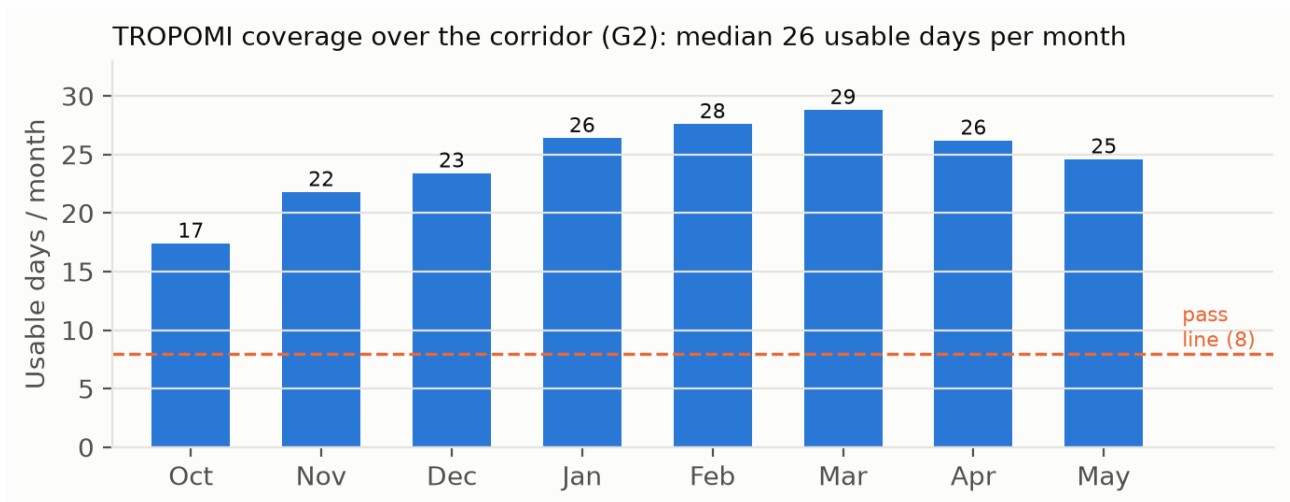


Figure 3. Usable TROPOMI overpasses per month (5-season average). October is weakest (late-monsoon cloud); every month clears the pass line comfortably.

4. Phase 1: measuring the city's emissions from space

Step 1, find the source. On calm days NO_2 piles up over its source. The calm-day average peaks over central Pune (Swargate/Camp), just south of the corridor.

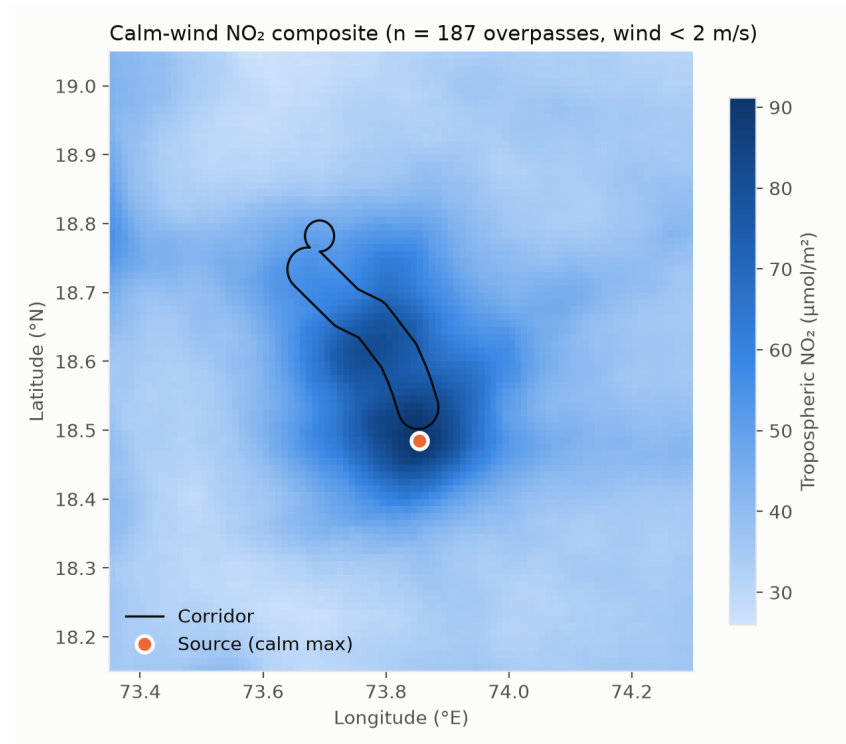


Figure 4. Average NO_2 on 187 calm days. Orange dot: the main source.

Step 2, how much? Each day's image is rotated so its wind points the same way, then averaged; summing across the wind gives the *line density*. A plume curve (EMG) is fitted, giving the NO_2 in the plume and how far it travels before decaying. Result: **0.663 kg/s NO_x** [95% CI 0.620–0.715], lifetime **1.09 h**.

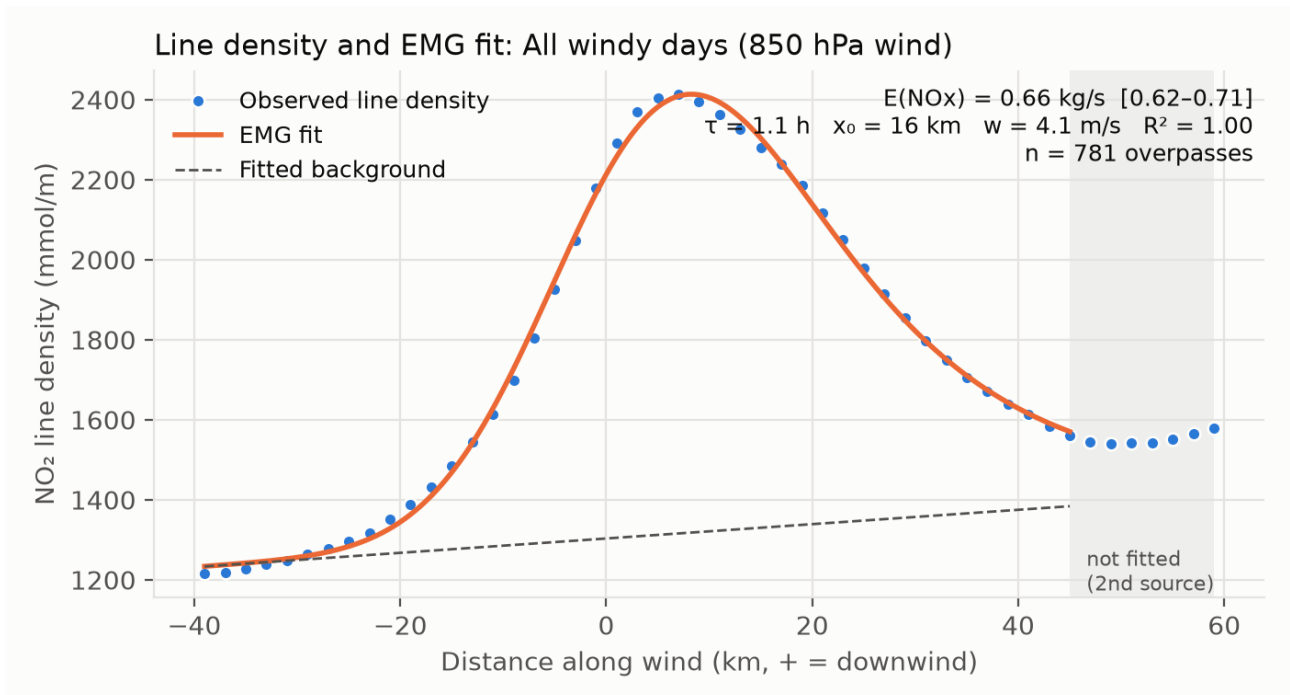


Figure 5. Line density (blue dots) and the fitted plume curve (orange), fitted up to 45 km downwind. The grey band is excluded because a second source sits there (see section 6).

Step 3, where exactly? A second method (flux divergence) maps emission pixel by pixel: what flows out minus what flows in, plus what chemistry destroys. It finds two hotspots 16.7 km apart: central Pune and Pimpri–Chinchwad. **The corridor produces about 21% of the metro's NO_x**, a share stable to $\pm 2.4\%$ in every test.

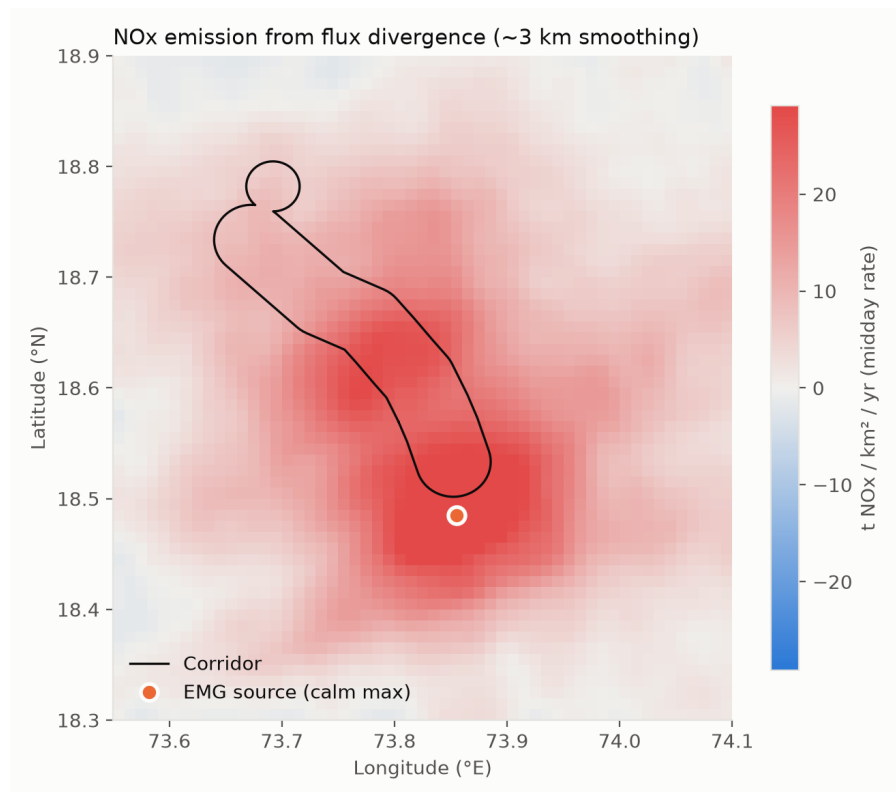


Figure 6. Emission map from flux divergence. Red = emitting.

5. From NOx to CO₂, and the inventories

NOx is turned into CO₂ with a CO₂:NOx ratio. Inventories say it varies by sector from 115 (industry) to 393 (households), so the city's mix matters. **TROPOMI CO measures that mix:** CO does not decay, so downwind of the city it steps up and stays up. After removing the Western Ghats terrain imprint, the step gives a CO:NOx of **16.7**, close to the inventory's 16.0, and narrows the CO₂:NOx ratio to **163 (148–180)**.

TROPOMI CO: downwind step over Pune (terrain-normalised, static pattern removed)

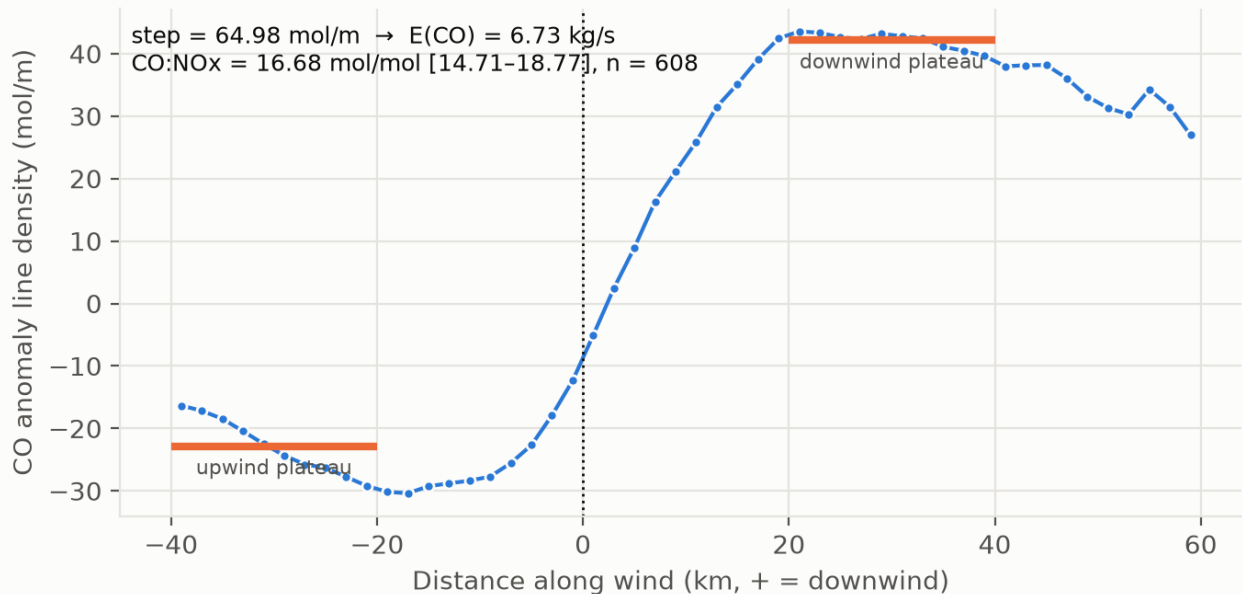


Figure 7. The CO 'step' over Pune: upwind plateau (left) to downwind plateau (right).

Uncertainty was propagated with a Monte Carlo (200,000 simulations) through every step, and the satellite's busy midday window was converted to an annual mean with published hour-by-hour emission profiles.

Monte Carlo uncertainty (200,000 draws): median, 68% (thick) and 95% (thin) ranges

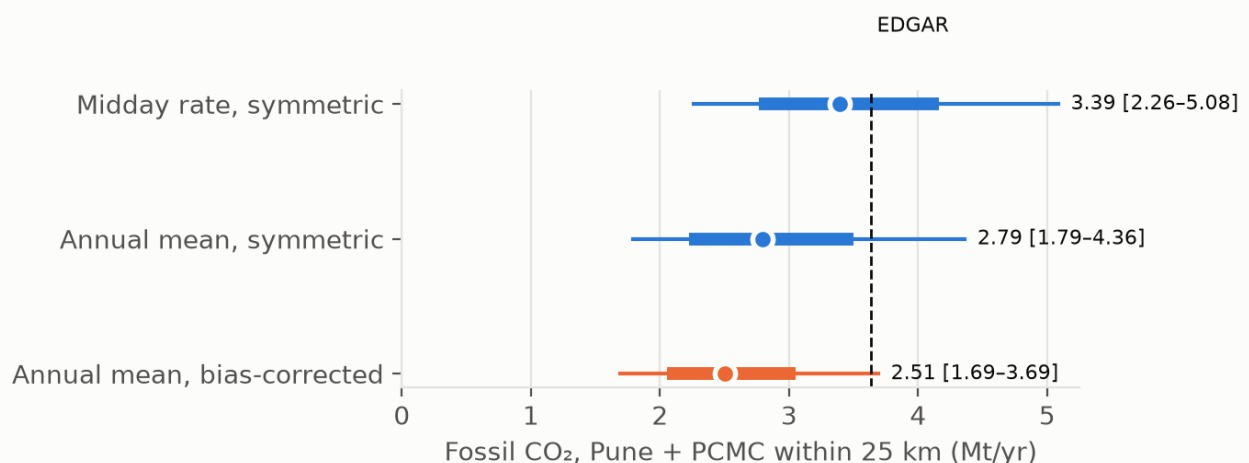


Figure 8. Fossil CO₂ for Pune + PCMC: median, 68% and 95% ranges. EDGAR (dashed) is inside; ODIAC (11.8, off the chart) is far outside.

Can CO₂ satellites confirm it? Not yet: Pune's CO₂ plume (~0.02–0.15 ppm) is 10–20× smaller than the noise and scan-stripe artefacts of OCO-3/OCO-2. The 8 snapshots are consistent with our estimate but only set an upper limit (about 7–14 Mt/yr). **So multi-satellite data help through CO, not through direct CO₂.**

OCO soundings (colour) and predicted Pune plume (contours 0.05 / 0.15 / 0.3 ppm)

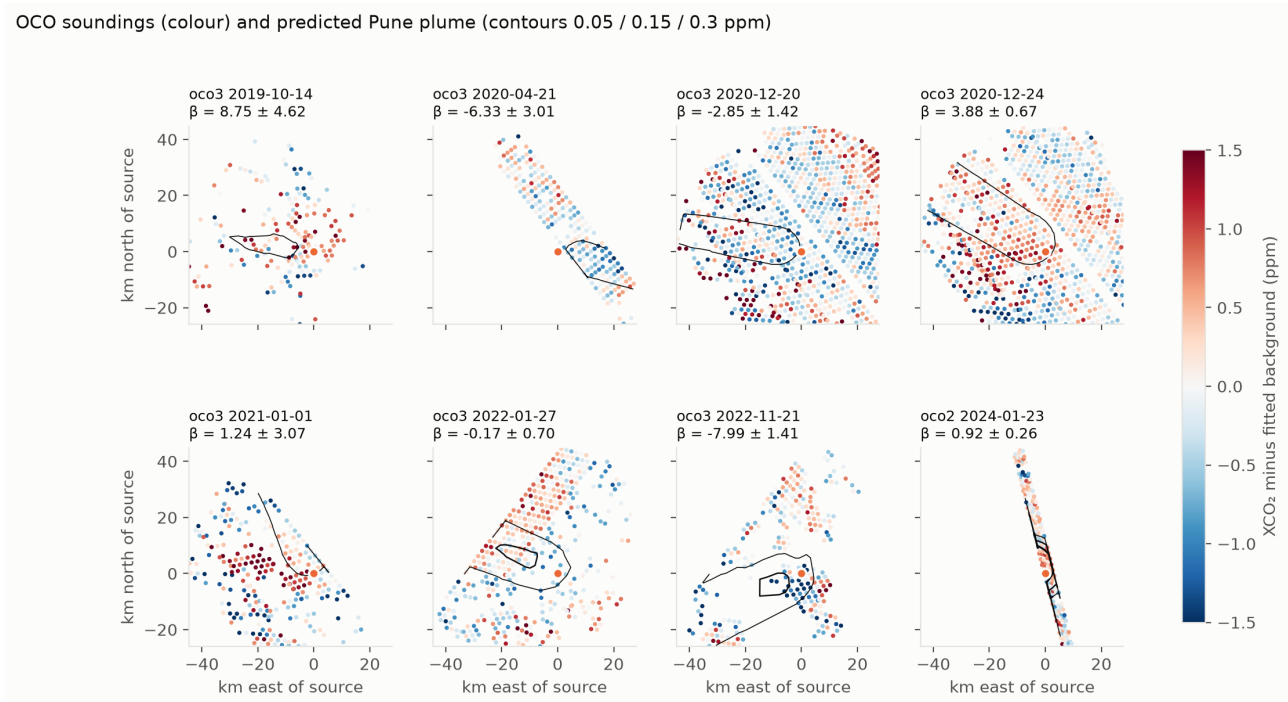


Figure 9. OCO soundings on the 8 dates (colour) vs the predicted Pune plume (contours). The stripes are instrument artefacts larger than the plume.

6. Testing the method, and a correction it caught

Every method was first run on **fake data with a known answer**: a simulated city on the real grid with real-like winds and noise. The plume fit recovers emissions within +12% (a measured, systematic bias kept in the uncertainty), flux divergence within -4%, the CO step within -1.4%.

A robustness test (letting the background slope) then changed the answer by 42%. Investigating showed the original fit reached 60 km downwind, where **PCMC and Talegaon add a second source** that the single-source model misread. Fitting only to 45 km fixed it: both background shapes agree, and NO_x rose ~20%. Everything downstream was rerun, and one earlier conclusion that depended on the low value ("more household burning than the inventory") was withdrawn.

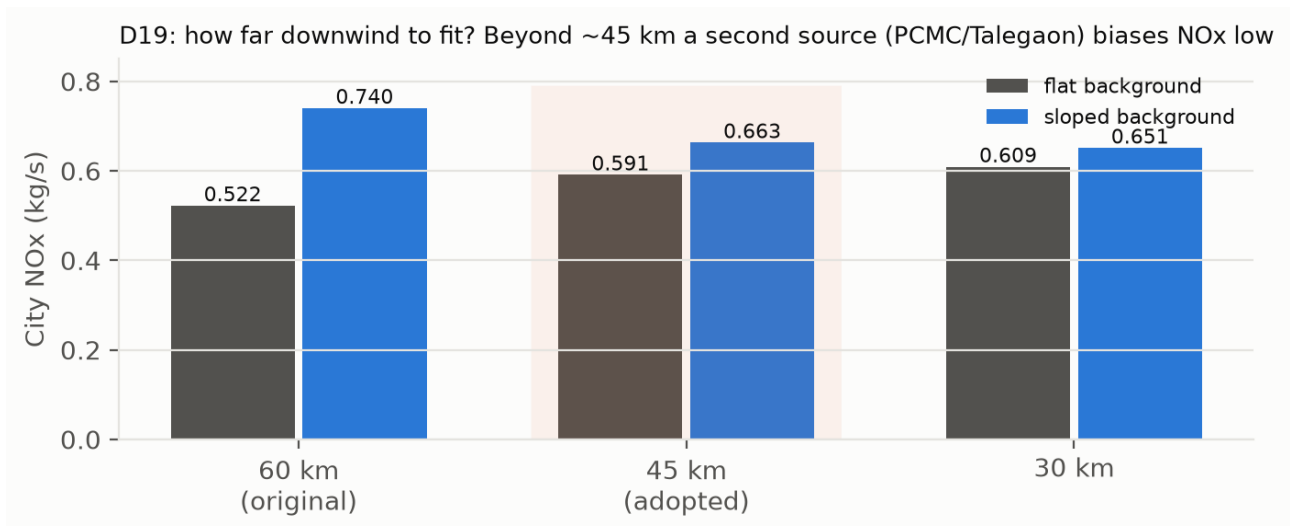


Figure 10. City NO_x by fit window and background shape. Inside 30–45 km the two agree (decision D19).

Does the method see real changes? A model-free check (city minus rural NO₂) for late March–May of each year:

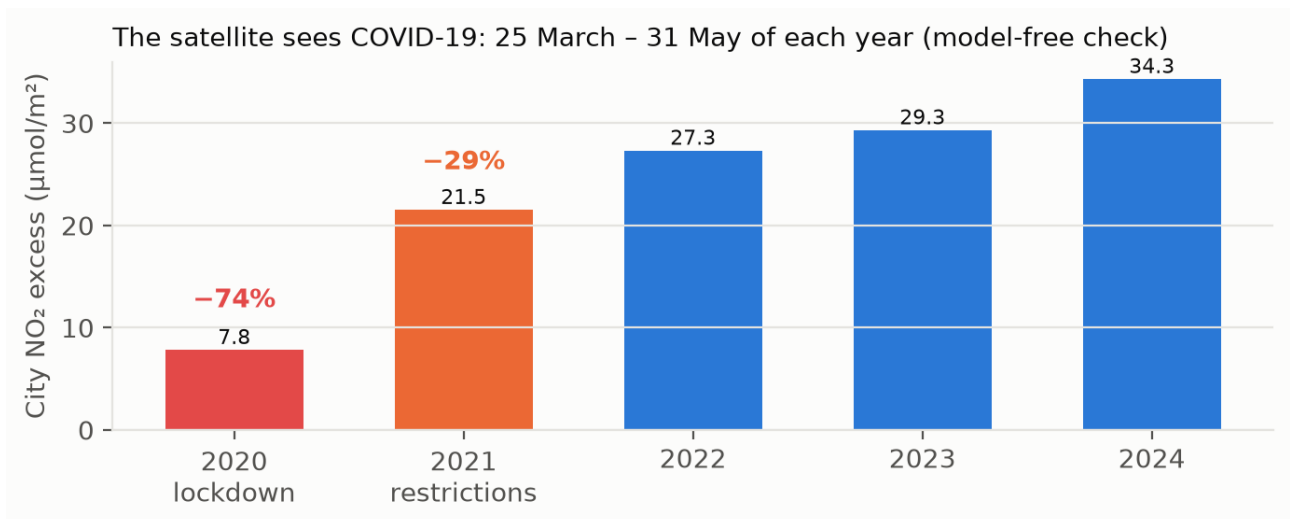


Figure 11. The 2020 national lockdown (-74%) and the 2021 Maharashtra restrictions (-29%) are both detected, in the right order, plus growth to 2024.

7. Phase 2: the 1 km feature table

For the machine-learning phases, every layer was put on the 1 km grid, one row per cell per month: 268 cells × 40 months = **10,720 rows, 17 features, 0 missing values** (feature table v2, frozen with a SHA-256 fingerprint). The features are NO₂, CO, HCHO (a chemistry proxy), wind and weather, night lights, vegetation and built-up indices, road density and industrial-land fraction. Inventories are kept only as reference columns, never as inputs.

Four of the 17 features on the 1 km grid (average over 40 months)

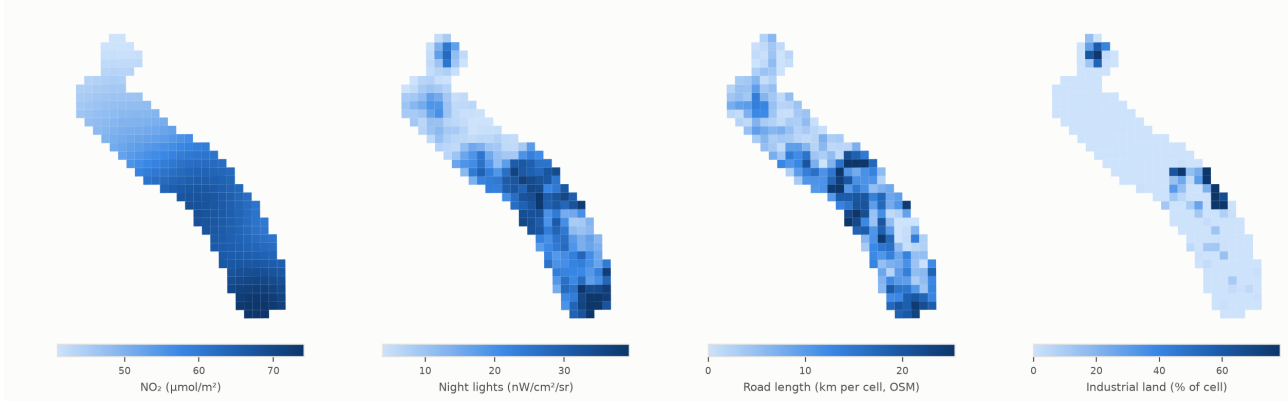


Figure 12. Four features averaged over 40 months. NO₂ is highest near Pune; night lights and roads trace the urban core and PCMC; industrial land picks out MIDC Bhosari, Chinchwad and Talegaon MIDC.

Roads and industry from OpenStreetMap. The OpenStreetMap query servers were overloaded for days, so the whole western-India map file (221 MB, checksum-verified) was downloaded once and read locally in 2.5 minutes. It gives 36,082 roads and 223 industrial areas. A second published road map (GRIP4) agrees on major roads (rank correlation 0.84) but misses almost all local streets (0.07 vs 4.2 km per cell), so OpenStreetMap is the better source.

A check before any machine learning: which inputs can say where? For each feature we measured how much of its variation is between cells rather than between months. Roads, industry, land cover and night lights vary in space; CO, HCHO and weather barely differ between 1 km cells. So at this scale the location of emissions must come from the activity layers (and partly NO₂), while CO and weather help with timing. This is written down *before* training, and it is why the models will be tested on held-out map areas (spatial cross-validation).

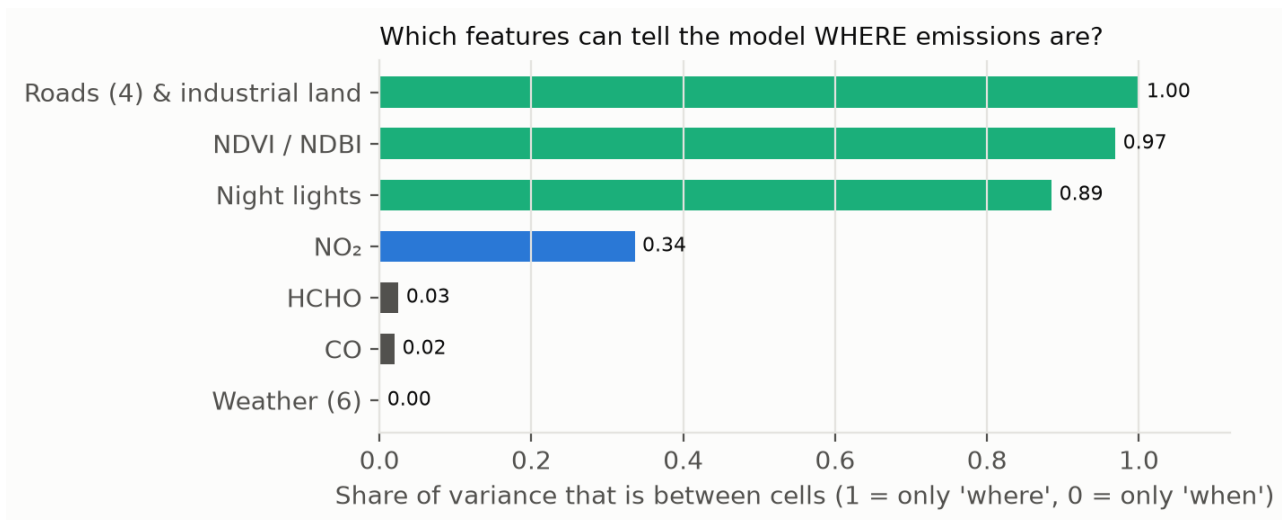


Figure 13. Share of each feature's variance that is spatial (between cells).

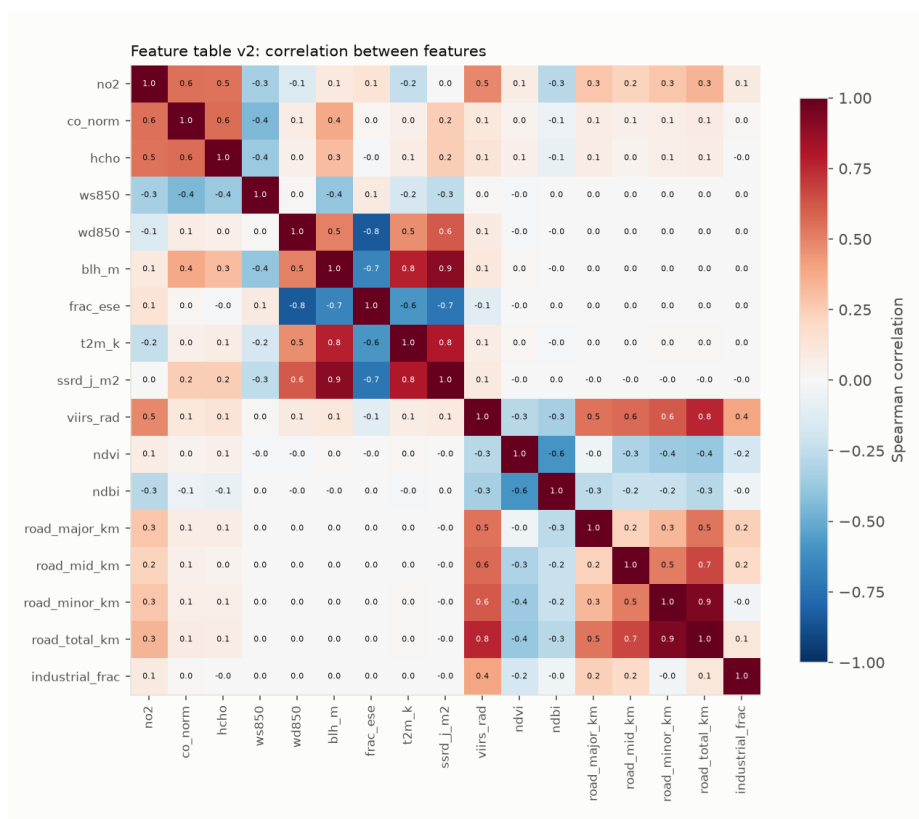


Figure 14. Correlation between features. Only expected pairs are strongly related (minor vs total roads; boundary layer, sunlight and temperature; two wind descriptors). Industrial land is nearly independent of everything.

8. Key decisions and lessons

Twenty design decisions are recorded with their evidence (docs/decisions.md). The most important:

#	Decision	Why
D1/D10	Measure the city total with the plume fit; split it by area with flux-divergence shares	Waypoints are closer than one satellite pixel; shares are robust
D9	Corridor built on the real highway + Talegaon MIDC	Proposal waypoints up to 4.4 km off
D11	OCO used as an upper limit, not a measurement	Pune's CO ₂ plume is below detection
D12	CO ₂ :NOx ratio constrained by TROPOMI CO	Largest error term; CO reveals the sector mix
D13	Outlier filter: each pixel vs its own history	Other filters clipped the plume or removed 29% of data
D17/D18	Annual-mean conversion; Monte Carlo uncertainty	Like-for-like comparison; honest ranges
D19	Fit only to 45 km downwind, sloped background	A second source biased NOx ~20% low
D14/D15/D20	HCHO as chemistry proxy; 1 km UTM grid; OSM roads via a bulk file (GRIP4 as fallback and cross-check)	Data availability and quality

Lessons that shaped the work

- Test every method on fake data with a known answer before trusting it, and use realistic conditions (Pune's real winds).
- Check outputs against physics: 439 'overpasses' in a month, 29% 'outliers' or a 2.6× unit mismatch were all bugs.
- When a method is fixed, rerun everything downstream and re-check every conclusion.
- Report null results honestly: OCO's non-detection is a finding, not a failure.
- When a server is the bottleneck, look for a bulk download of the same data (OpenStreetMap: days of failures vs 2.5 minutes).
- Write everything down: a research log, a decision record and a findings document were kept throughout.

9. Where the project stands, and what comes next

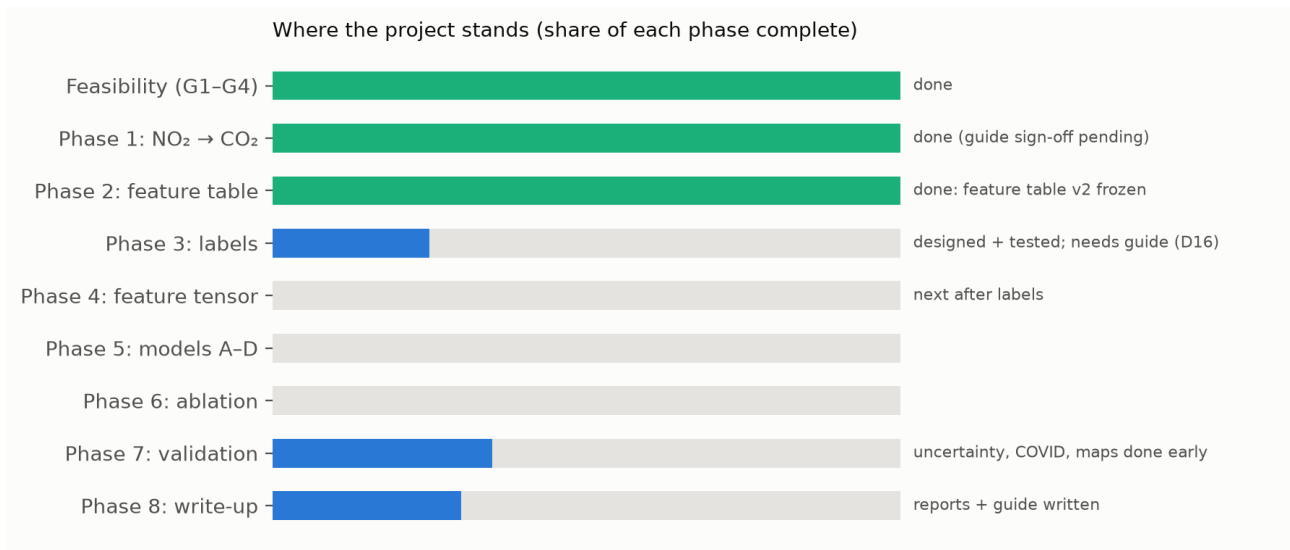


Figure 15. Progress by phase.

Next steps

- **Guide review and sign-off** of the open decisions: OCO's role (D8/D11), the CO-constrained ratio (D12), the fit-window correction (D19) and the Phase 3 label design (D16).
- **Phase 3, labels:** CO₂ per cell and season = city total × flux-divergence share; test a CO-based label that is independent of NO₂, so the experiments are not circular.
- **Phases 4–6, machine learning:** train tree models for Experiments A–D (NO₂ only → + CO → + weather → + human activity), with spatial-block cross-validation and an ablation, to measure what each data source adds.
- **Phase 7–8:** validation maps, final uncertainty, and the final report.

Questions for the guide: (1) Is the seasonal flux-divergence label acceptable, given its link to NO₂ is stated? (2) Should the CO-based label become a second label? (3) Is seasonal resolution (1,340 labelled samples) enough? (4) Should Experiment B be redefined around TROPOMI CO, since OCO cannot be a per-cell feature?

Full detail: [docs/phase1_report.md](#), [docs/findings.md](#), [docs/decisions.md](#), [docs/phase3_design.md](#), and the learning guide in [docs/guide/](#).